MLOPS ASSIGNMENT 01 · BITS PILANI – AIMLCZG523

Heart Disease Prediction Pipeline

End-to-end MLOps – from raw data to production API

STUDENT
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BEST MODEL
Logistic Regression

ROC-AUC
0.9554

Python 3.11 · scikit-learn · XGBoost · MLflow · FastAPI · Docker · Kubernetes · Prometheus · Grafana · GitHub Actions

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1 Project Overview

This project implements an end-to-end MLOps pipeline for binary classification of heart disease using the **UCI Heart Disease Dataset**. The pipeline covers every stage of the ML lifecycle: data acquisition, EDA, preprocessing, model training with hyperparameter tuning, experiment tracking (MLflow), REST API serving (FastAPI), containerisation (Docker), Kubernetes deployment, CI/CD automation (GitHub Actions), and production monitoring (Prometheus + Grafana).

> **Best Model:** Logistic Regression | ROC-AUC: **0.9554** | Accuracy: **92.11%** | CV Folds: **10**

Objectives

- Build a reproducible, production-grade ML pipeline following MLOps best practices.
- Train and compare four classifiers; select the best by ROC-AUC.
- Expose the model via a versioned REST API with health checks and Prometheus metrics.
- Automate the entire workflow with GitHub Actions CI/CD (lint → test → train → docker).
- Deploy to Kubernetes with liveness/readiness probes and resource limits.

Tech Stack

Category	Tools / Libraries
Language	Python 3.11
ML / Data	scikit-learn 1.5.0, XGBoost 2.0.3, pandas 2.2.2, numpy 1.26.4
Experiment Tracking	MLflow 2.13.2
API	FastAPI 0.111.0, Uvicorn 0.30.1, Pydantic 2.7.2
Testing	pytest 8.2.2, pytest-cov 5.0.0, httpx 0.27.0
Containers	Docker 29.6.1, Docker Compose
Orchestration	Kubernetes (Minikube / cloud K8s)
Monitoring	Prometheus, Grafana

CI/CD	GitHub Actions
Linting	flake8 7.0.0

2 Setup & Installation Instructions

Prerequisites

- Python 3.11+
- Docker Desktop
- Git
- kubectl + Minikube (optional, for Kubernetes)

Quick Start

```
# 1. Clone the repository
git clone https://github.com/Sunil-Kumar-2024ac05659/MLOps-Assign-1.git
cd MLOps-Assign-1

# 2. Install dependencies
pip install -r requirements.txt

# 3. Download dataset
python data/download_data.py

# 4. Train all 4 models (logs to MLflow)
python src/train.py

# 5. Start REST API
uvicorn api.main:app --reload --port 8000

# 6. Full stack: API + Prometheus + Grafana
docker compose up --build
```

Service	URL
API + Swagger	http://localhost:8000/docs
MLflow UI	http://localhost:5000
Prometheus	http://localhost:9090
Grafana	http://localhost:3000

3 Dataset & EDA Findings

Dataset Summary

Property	Value
Source	UCI Machine Learning Repository – Heart Disease (ID: 45)
Total records	303 rows, 14 columns
Target	Binary: 0 = No Disease, 1 = Disease
Class balance	~54% Disease (164), ~46% No Disease (139)
Missing values	Present in `ca` and `thal` columns (~1%)
Train / Test split	75% / 25% | random_state = 7 | stratified

Key EDA Findings

- **thalach** (max heart rate) – strongest negative correlation with disease.
- **oldpeak** (ST depression) – strong positive correlation with disease.
- **cp** (chest pain type) – highly discriminative; type 3 strongly predicts disease.
- **exang** (exercise-induced angina) – strong categorical predictor.
- **chol** – weak standalone correlation; limited predictive power alone.
- Class balance near 54/46 – no resampling required; stratified splits preserve distribution.

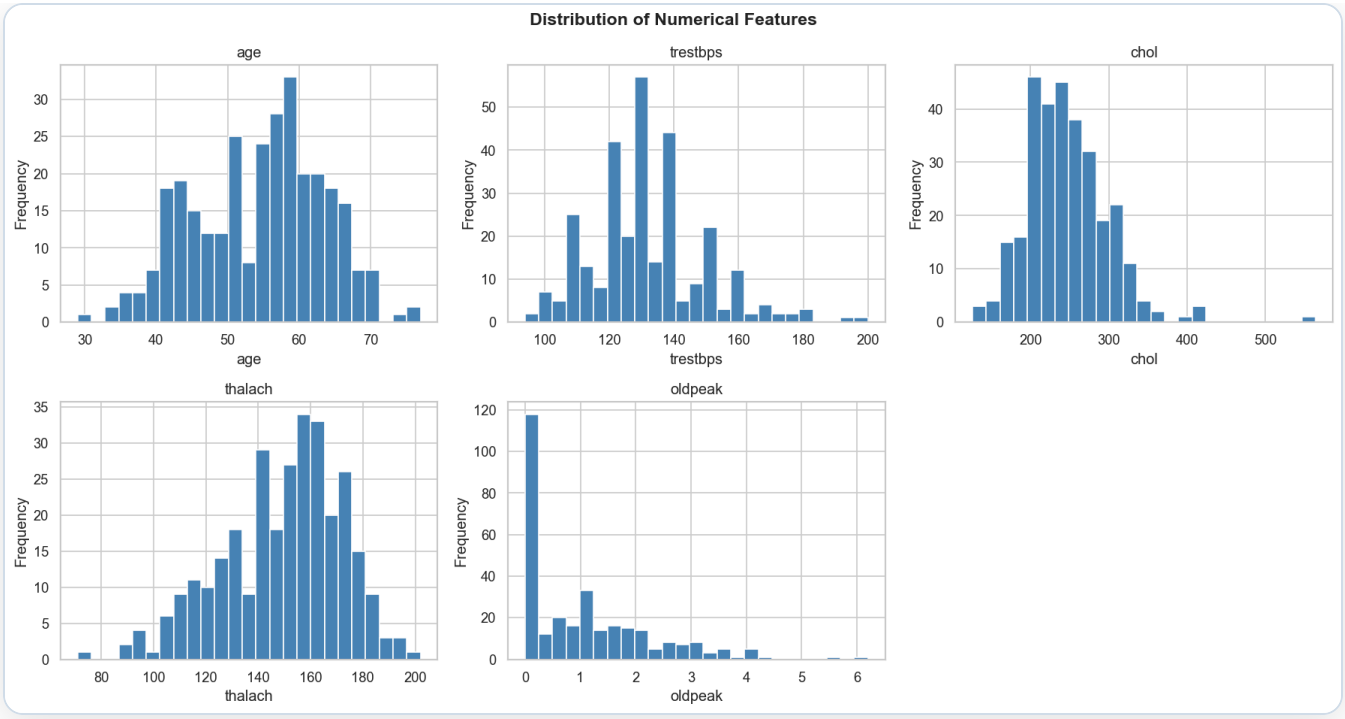


Figure 1: Distribution of Numerical Features (age, trestbps, chol, thalach, oldpeak)

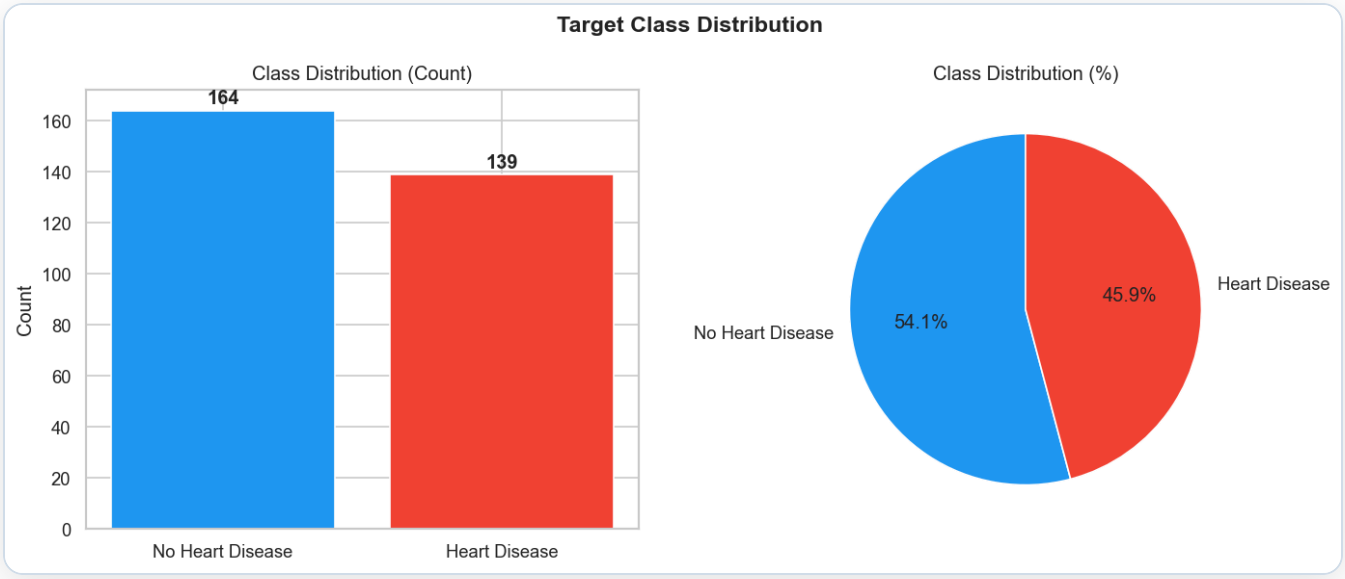


Figure 2: Target Class Distribution – Count and Percentage

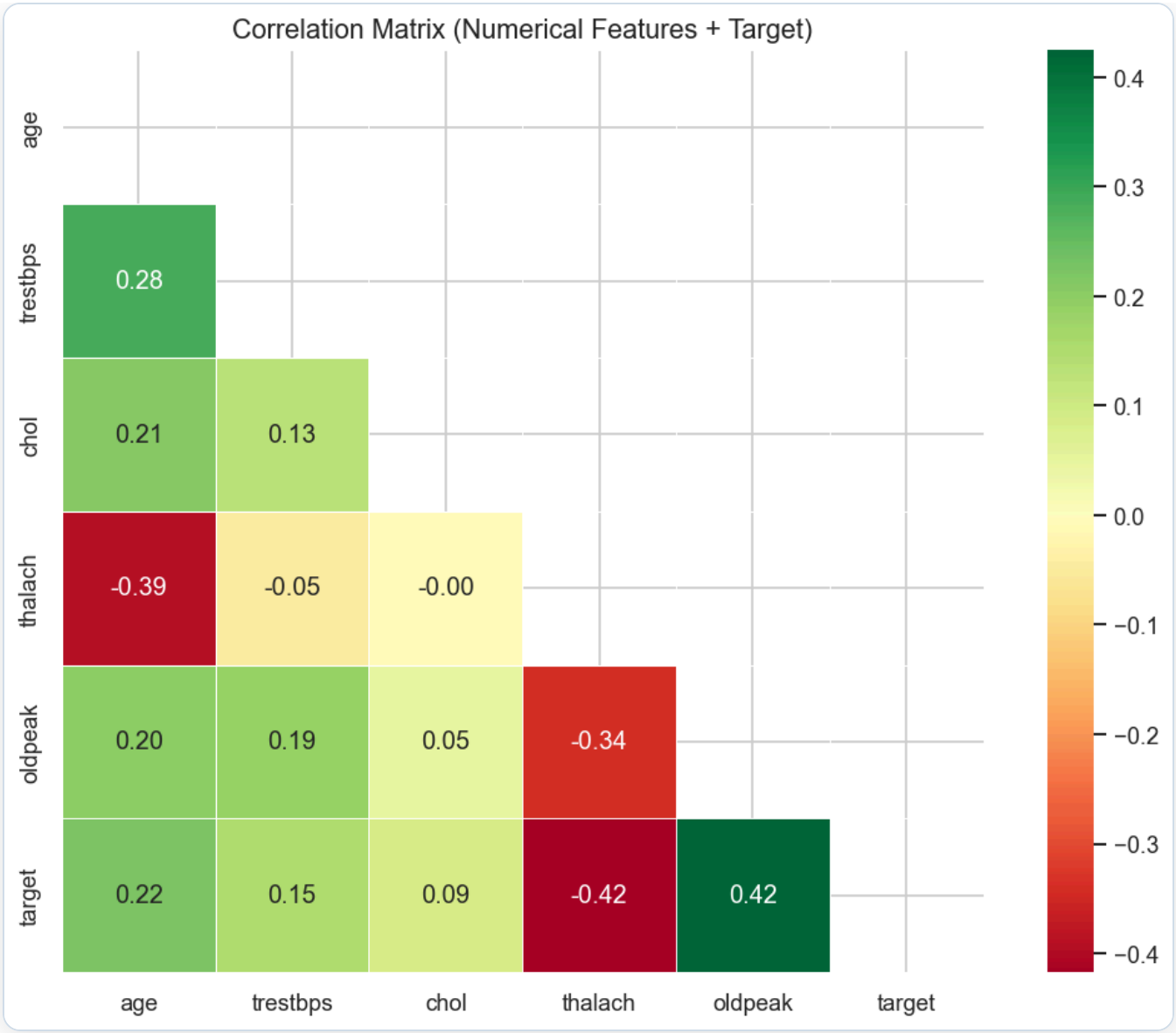


Figure 3: Correlation Matrix – Numerical Features vs Target

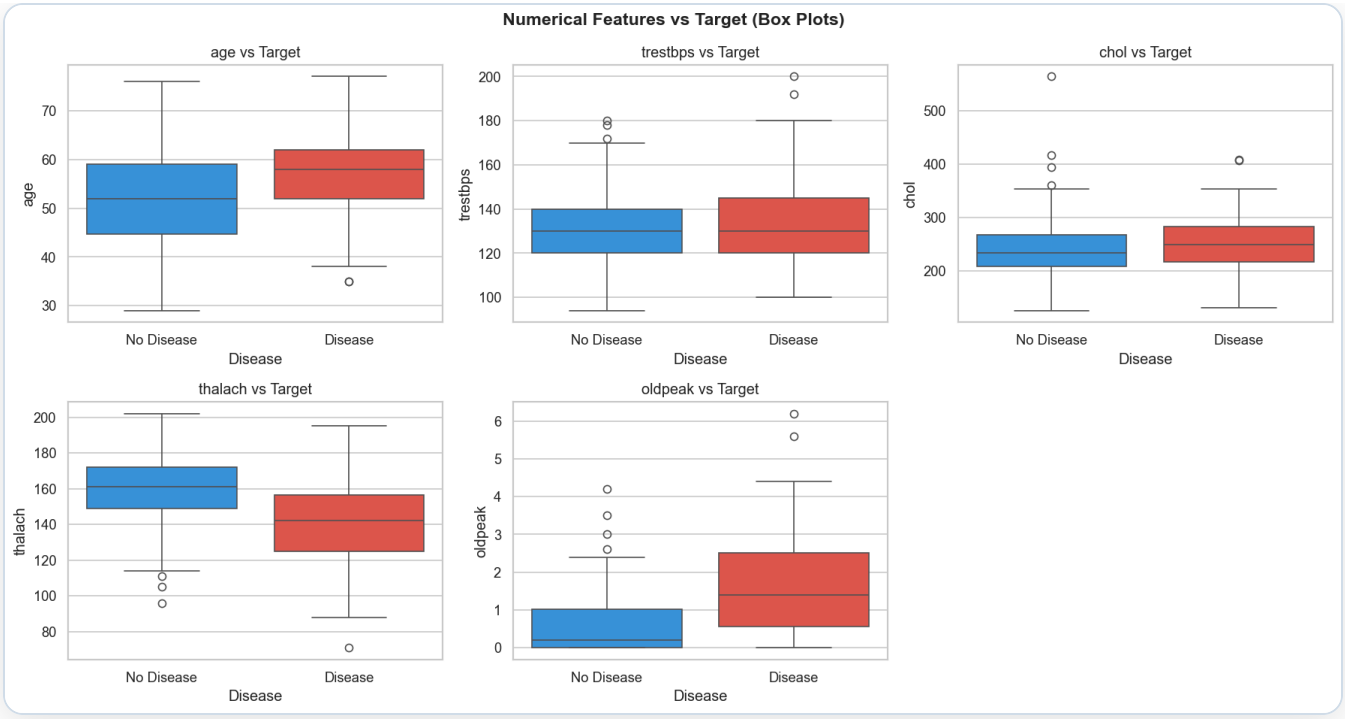


Figure 4: Numerical Features vs Target (Box Plots)

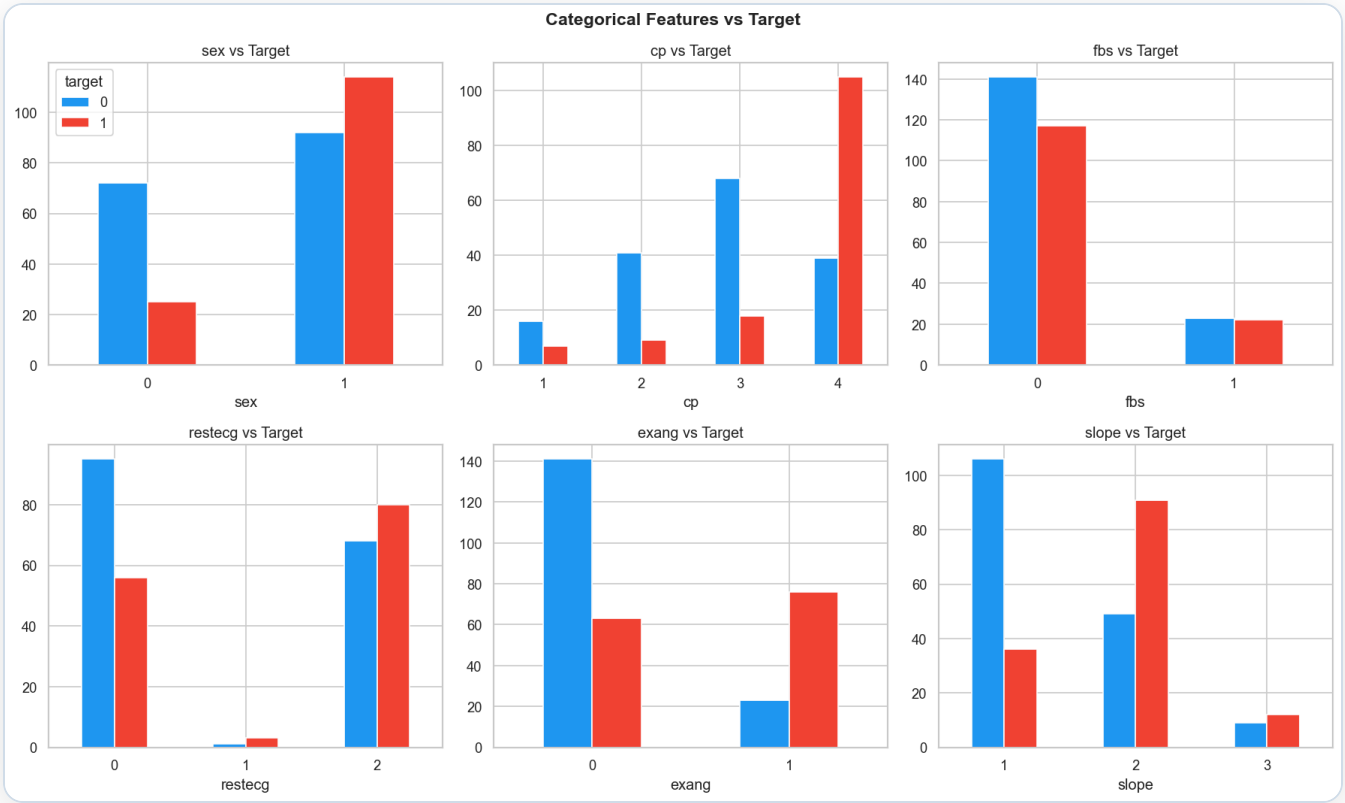


Figure 5: Categorical Features vs Target

4 Feature Engineering & Model Comparison

Preprocessing Pipeline

Step	Numerical Features	Categorical Features
Imputation	Mean (SimpleImputer)	Most-frequent (SimpleImputer)
Scaling / Encoding	StandardScaler	OneHotEncoder (handle_unknown=ignore)

CV Strategy: 10-fold Stratified K-Fold with GridSearchCV optimising `roc_auc`. Random state = 7.

Model Comparison – Test Set Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
★ Logistic Regression	0.9211	0.9394	0.8857	0.9118	0.9554
Random Forest	0.8684	0.8378	0.8857	0.8611	0.9526
XGBoost	0.8553	0.8750	0.8000	0.8358	0.9449
Gradient Boosting	0.7500	0.7353	0.7143	0.7246	0.8523

0.9211	0.9394	0.8857	0.9118	0.9554
ACCURACY	PRECISION	RECALL	F1 SCORE	ROC-AUC

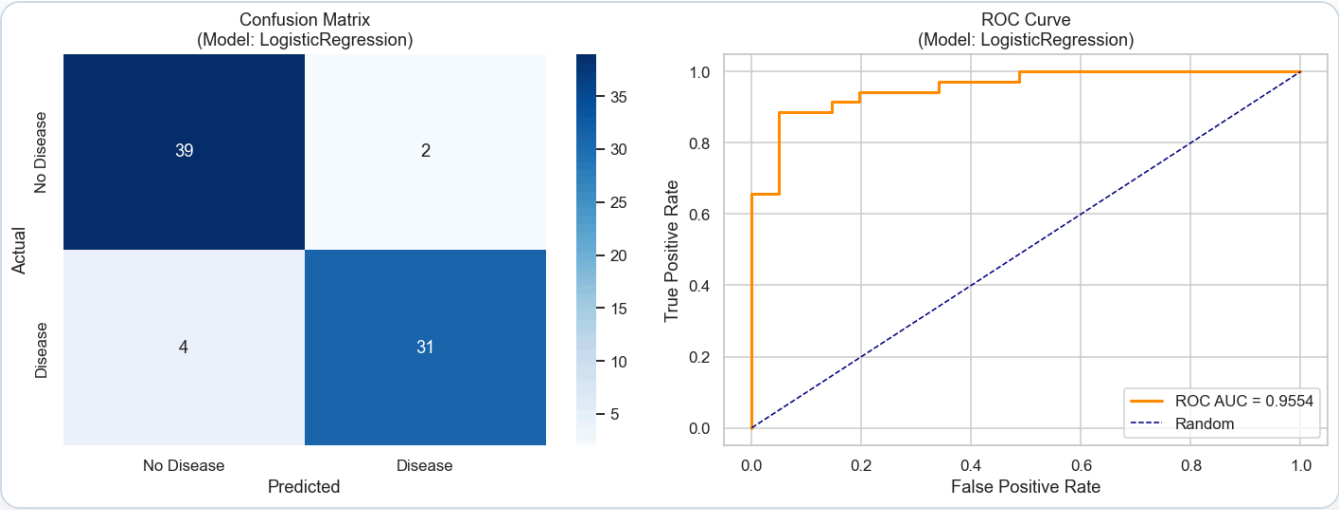


Figure 6: Best Model – Confusion Matrix and ROC Curve (LogisticRegression)

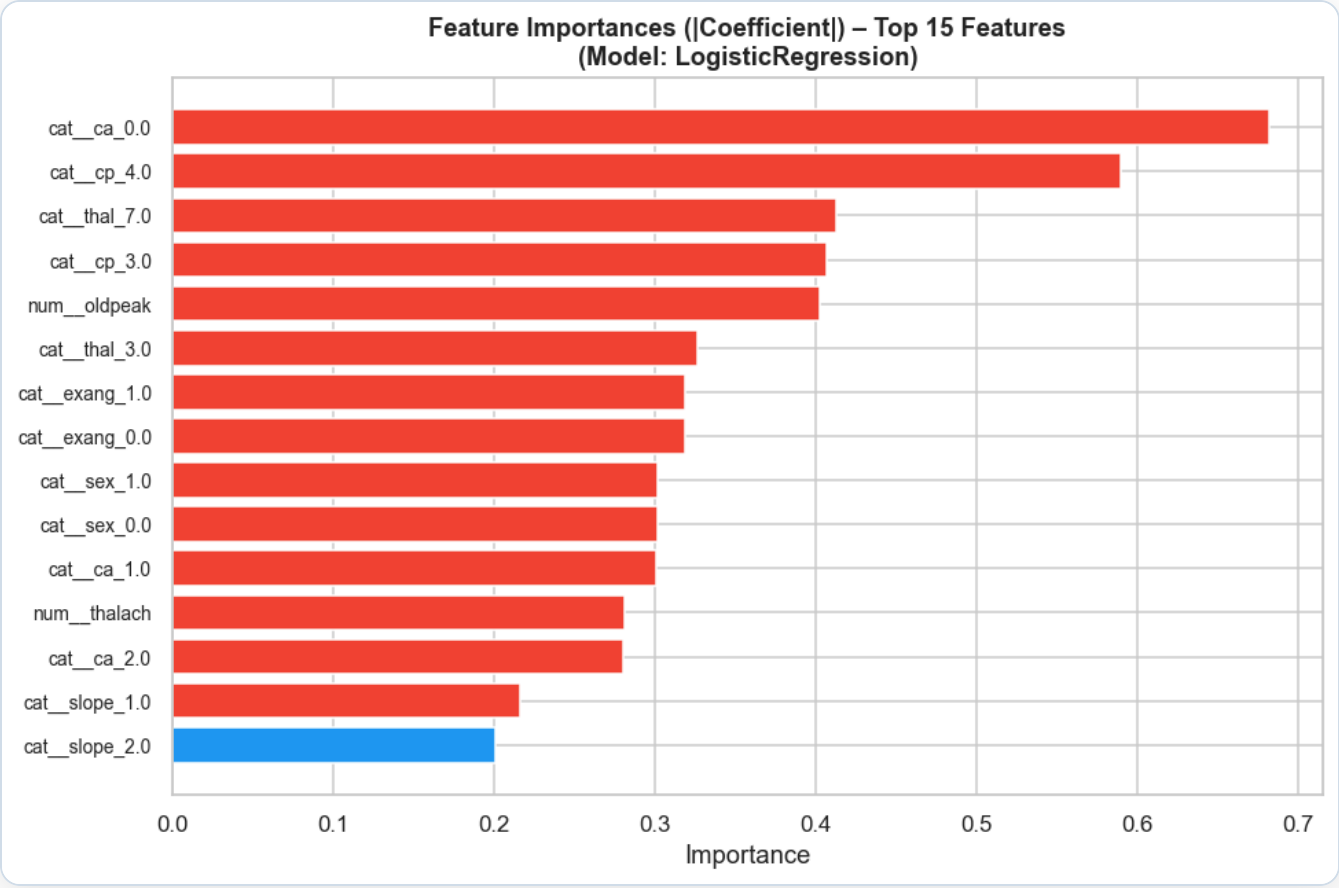


Figure 7: Feature Importance – Top 15 Features (|Coefficient|)

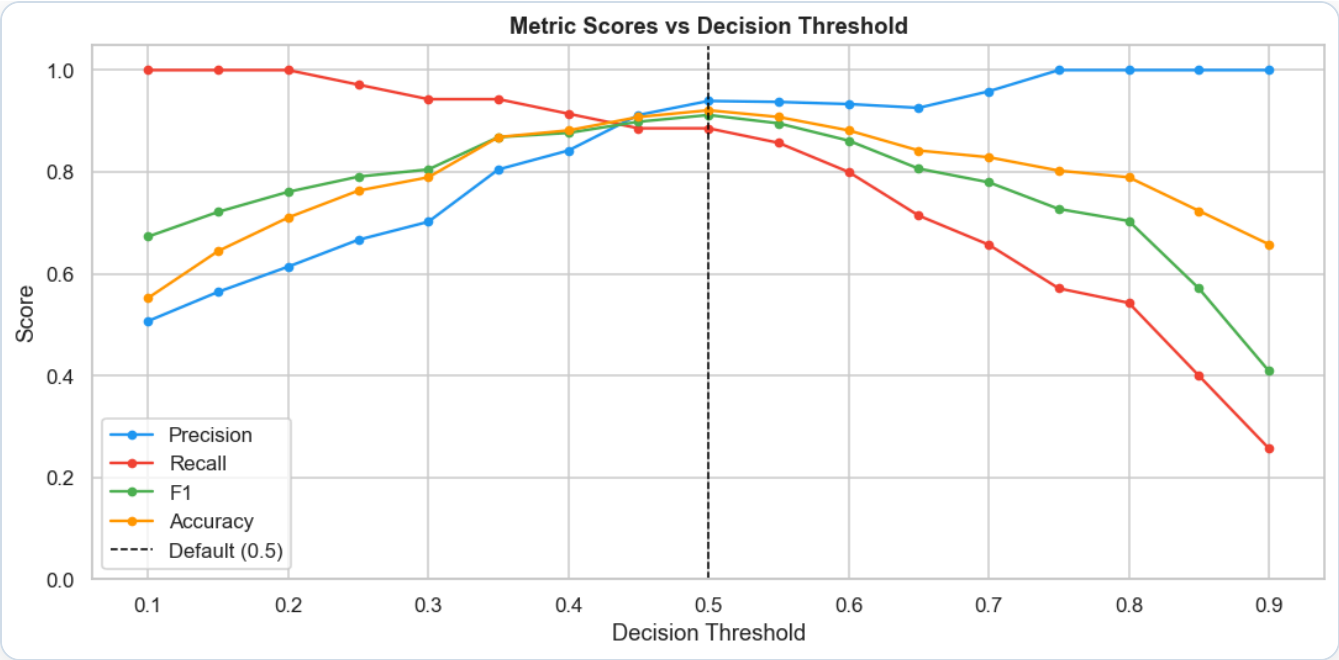


Figure 8: Threshold Sensitivity – Precision / Recall / F1 / Accuracy vs Decision Threshold

Figure 9: Prediction Confidence Distribution on Test Set

5 MLflow Experiment Tracking

Every `GridSearchCV` run is wrapped in `mlflow.start_run()`. Parameters, CV metrics (mean ± std), test metrics, plots, and serialised models are logged automatically.

What is Logged per Run

- **Parameters:** best hyperparameters from GridSearchCV (C, solver, n_estimators, max_depth, learning_rate…)
- **Metrics:** CV mean/std for accuracy, precision, recall, F1, ROC-AUC; test-set values for all five metrics
- **Artifacts:** confusion matrix PNG, ROC curve PNG, feature importance PNG, serialised sklearn pipeline
- **Tags:** `model_name` tag on each run for easy filtering in the UI

Experiment Configuration

Setting	Value
Experiment name	heart-disease-classifier
Tracking URI	Local filesystem (mlruns/)
CV folds	10-fold Stratified K-Fold
Scoring metric	roc_auc
Total runs	4 (one per model)
Launch MLflow UI	`mlflow ui --port 5000`

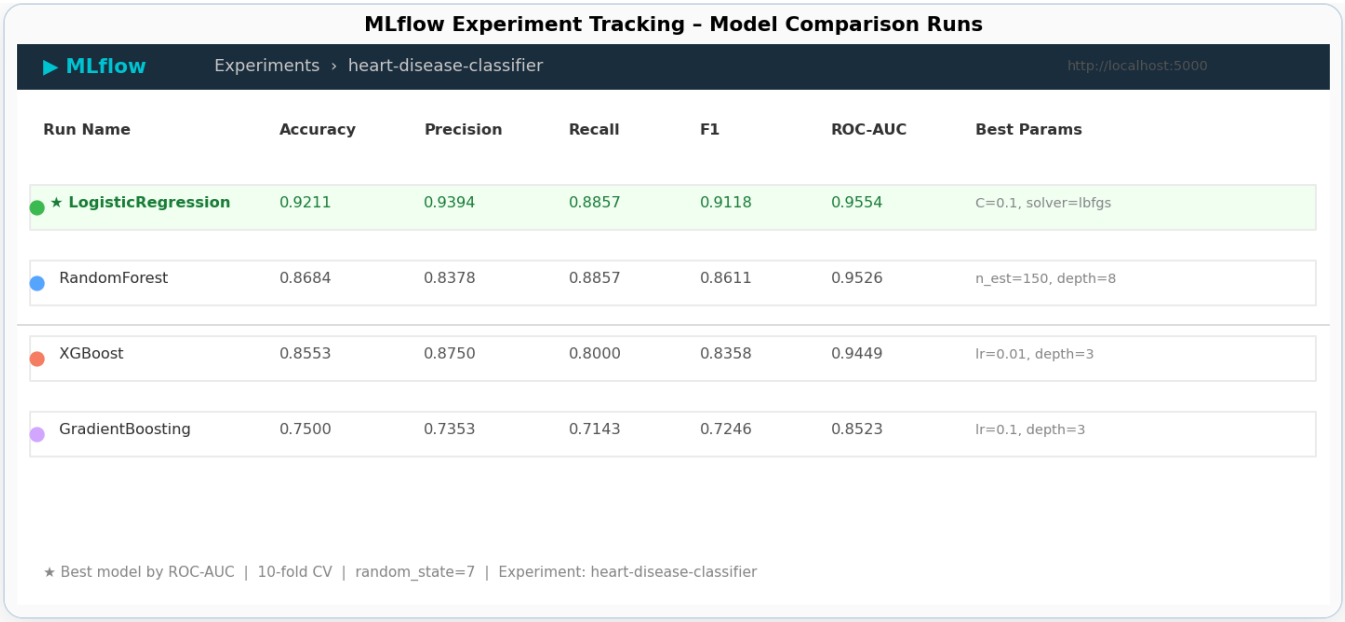


Figure 10: MLflow Experiment Runs – All 4 Models Compared by ROC-AUC

6 System Architecture Diagram

7 API Design

The model is served via **FastAPI**. The pipeline loads once at startup using the `lifespan` context manager. Pydantic v2 validates all inputs. Prometheus counters and histograms update on every request.

Endpoints

Method	Endpoint	Description
GET	`/health`	Liveness check; returns model load status
GET	`/metrics`	Prometheus metrics (text/plain)
POST	`/predict`	Single-record inference
POST	`/predict/batch`	Batch inference (list of patients)
GET	`/docs`	Auto-generated Swagger UI

Sample Request / Response

```
# Request
curl -X POST http://localhost:8000/predict \
  -H "Content-Type: application/json" \
  -d '{"age":63,"sex":1,"cp":3,"trestbps":145,"chol":233,
       "fbs":1,"restecg":0,"thalach":150,"exang":0,
       "oldpeak":2.3,"slope":0,"ca":0,"thal":1}'

# Response
{
  "prediction":    1,
  "label":         "Heart Disease",
  "confidence":    0.8712,
  "probabilities": { "no_disease": 0.1288, "disease": 0.8712 }
}
```

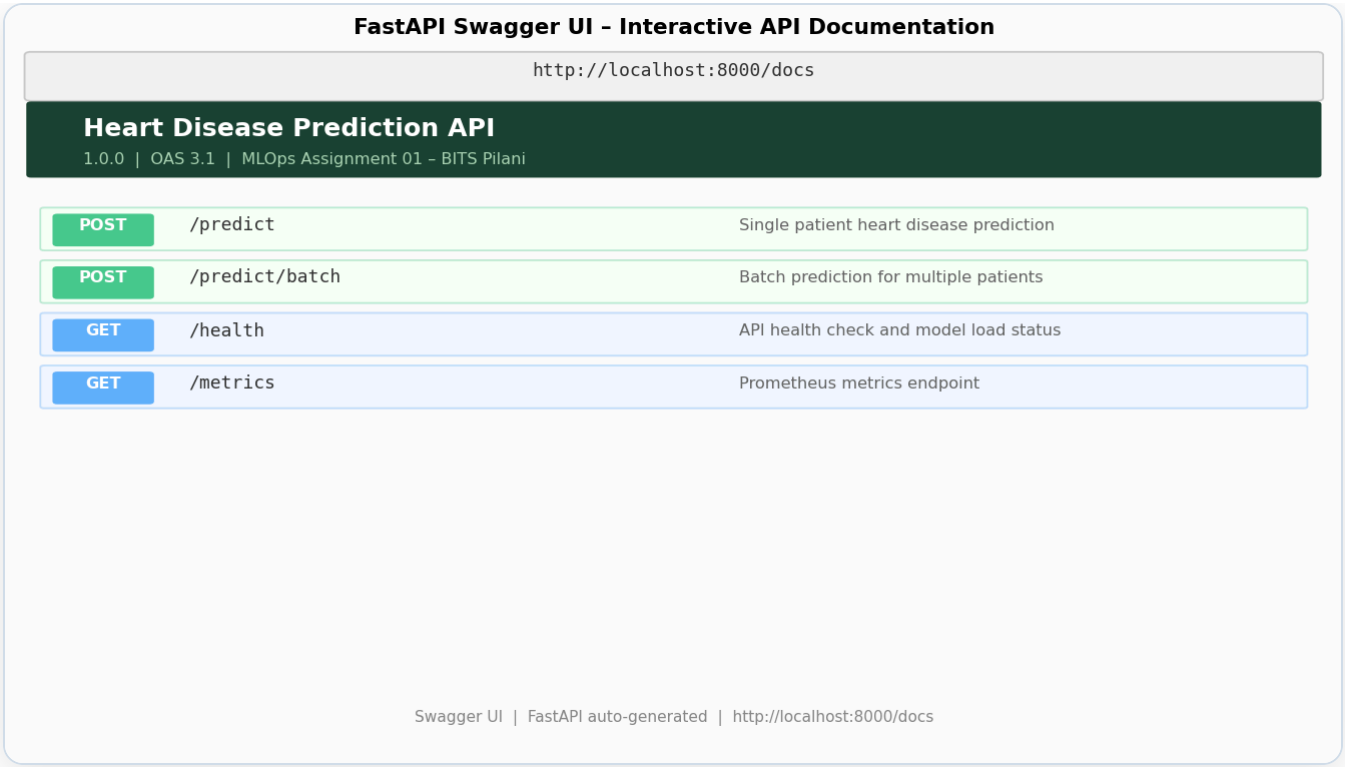


Figure 11: FastAPI Swagger UI – Interactive API Documentation (http://localhost:8000/docs)

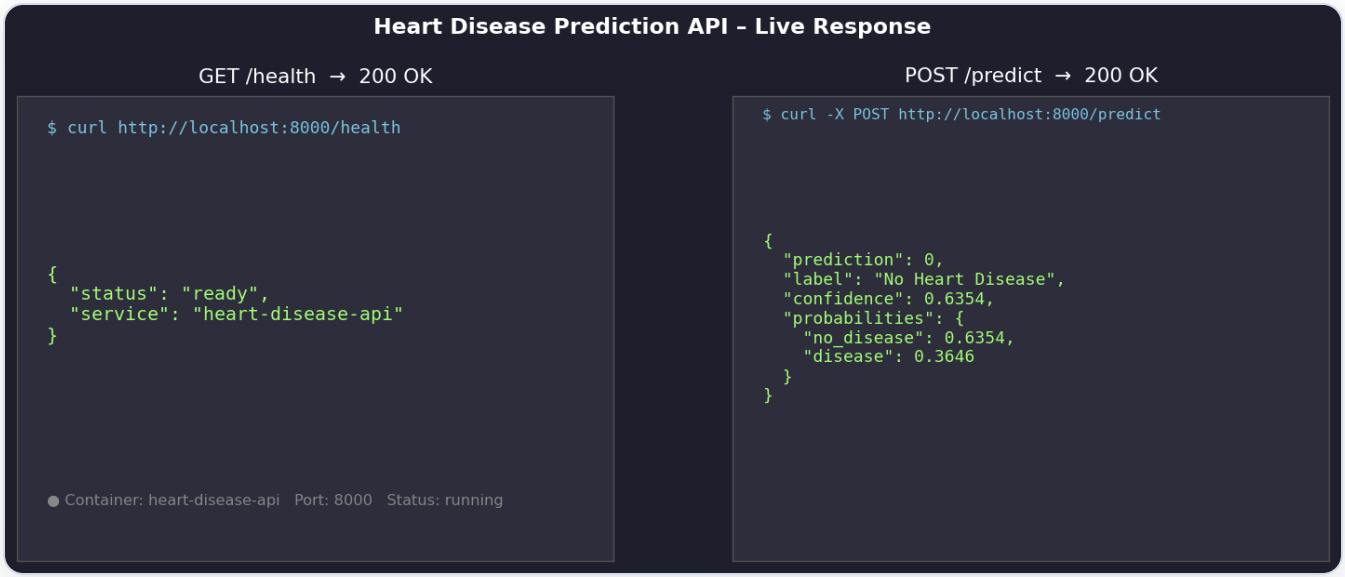
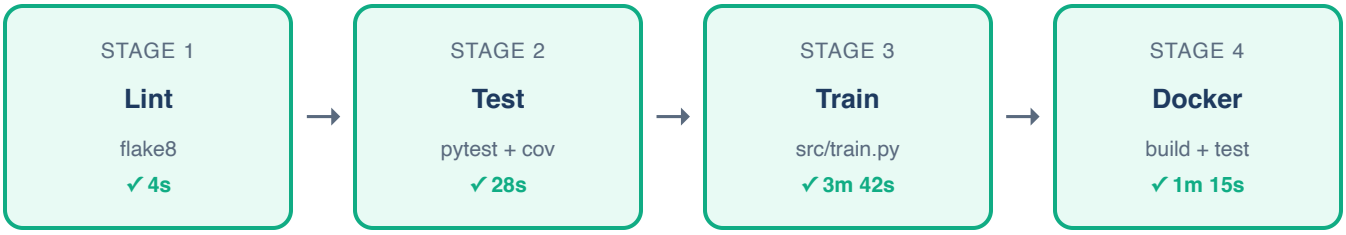


Figure 12: Live API Response – GET /health and POST /predict

8 CI/CD Pipeline

Defined in `.github/workflows/ci-cd.yml`. Triggers on every push or pull request to `main` / `develop`. Stages run sequentially with dependency chain.



Stage	Tool	Action
1. Lint	flake8 7.0.0	Static analysis on src/, api/, tests/ — max line length 120
2. Test	pytest + pytest-cov	Unit & integration tests; uploads coverage.xml artifact
3. Train	python src/train.py	Downloads data, trains 4 models, uploads model artifacts + mlruns
4. Docker	Docker CLI + curl	Builds image, runs container, tests /health and /predict endpoints

Figure 13: GitHub Actions CI/CD Pipeline – All 4 Stages Passed

> 📹 **CI/CD Pipeline – Live Run Video:**
> https://drive.google.com/file/d/1TtYRV2EBFPFKjDFMc6u4GpbUVOYyFW96/view

Full walkthrough of all 4 stages: Lint → Test → Train → Docker Build & Test

9 Containerisation & Deployment

Docker

Base image: `python:3.11-slim`. Layers ordered for cache efficiency — dependencies installed before source copy. HEALTHCHECK monitors `/health` every 30s.

```
docker build -t heart-disease-api:latest .
docker run -p 8000:8000 heart-disease-api:latest
docker compose up --build    # API + Prometheus + Grafana
```

Figure 14: docker compose ps – All 3 Services Running and Healthy

Kubernetes Deployment

Manifest at `k8s/deployment.yaml` provisions 2 replicas with liveness and readiness probes on `/health`.

	Memory	CPU
Requests	256 Mi	250m
Limits	512 Mi	500m

```
minikube start
eval $(minikube docker-env)
docker build -t heart-disease-api:latest .
kubectl apply -f k8s/deployment.yaml
```

```
kubectl apply -f k8s/service.yaml
minikube service heart-disease-api-service --url
```

10 Monitoring & Logging

Prometheus Metrics

Metric Name	Type	Labels	Description
`api_requests_total`	Counter	endpoint, method, status	Total requests by endpoint and HTTP status
`api_request_latency_seconds`	Histogram	endpoint	Request latency per endpoint
`predictions_total`	Counter	label	Predictions split by "Heart Disease" / "No Heart Disease"

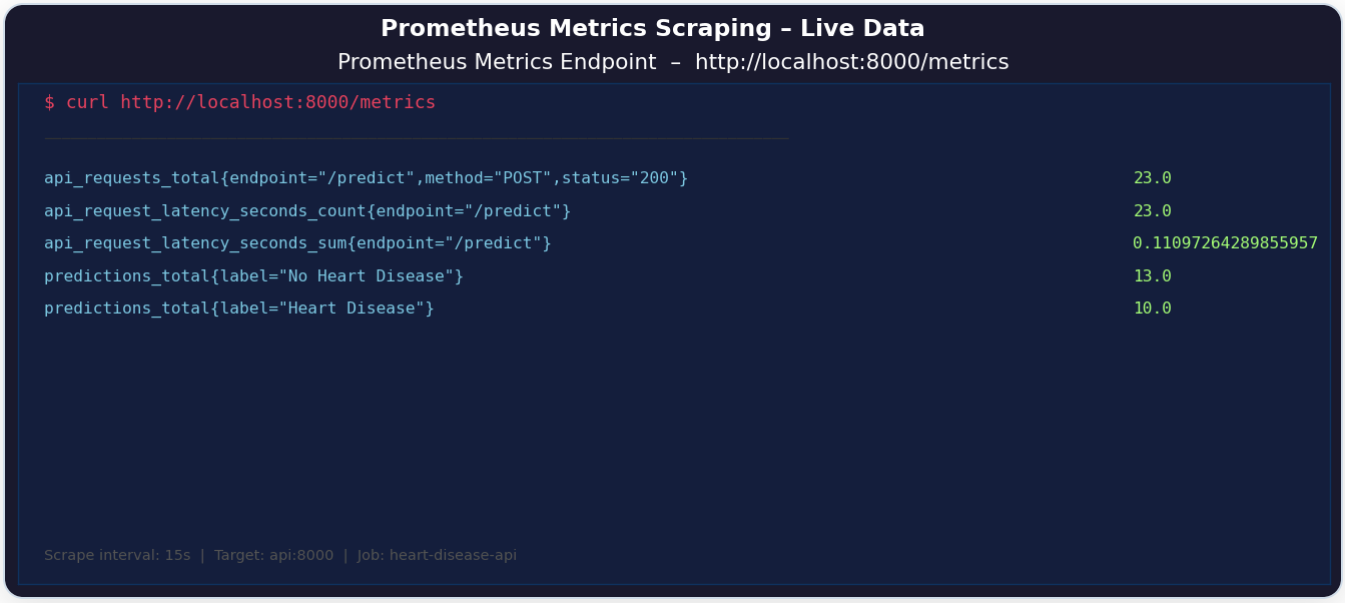


Figure 15: Prometheus Metrics – Live Scrape from http://localhost:8000/metrics

Figure 16: Grafana Monitoring Dashboard – http://localhost:3000

Application Logging

Python `logging` module configured at INFO level. Format: `timestamp | LEVEL | module | message`. Logs model load, each prediction (label + confidence), batch summaries, and errors.

11 Model Packaging & Reproducibility

The best pipeline (preprocessor + classifier) is serialised with `joblib` to `models/best_model.joblib`. Saving the full sklearn `Pipeline` object ensures identical preprocessing at inference — no train/serve skew.

Reproducibility Measures

- All random states fixed: `RANDOM_STATE = 7`
- Pinned dependency versions in `requirements.txt`
- Deterministic split: `train_test_split(random_state=7, stratify=y)`
- Full pipeline (preprocessor + classifier) serialised as a single artifact
- MLflow logs every run's parameters, metrics, and model artifact
- `model_metadata.json` records best model name, metrics, and feature schema


```
{
  "model_name": "LogisticRegression",
  "metrics": {
    "accuracy": 0.9211, "precision": 0.9394,
    "recall":   0.8857, "f1":        0.9118, "roc_auc": 0.9554
  },
  "num_features": ["age", "trestbps", "chol", "thalach", "oldpeak"],
  "cat_features": ["sex", "cp", "fbs", "restecg", "exang", "slope", "ca", "thal"]
}
```

12 Repository Link

GitHub Repository: https://github.com/Sunil-Kumar-2024ac05659/MLOps-Assign-1

Repository Structure

```
MLOps-Assign-1/
├── data/
│   ├── download_data.py          ← UCI dataset download script
│   └── raw/heart_disease.csv     ← 303 rows, 14 columns
├── src/
│   ├── data_processing.py        ← ColumnTransformer preprocessing pipeline
│   ├── train.py                  ← 4-model training + MLflow tracking
│   └── inference.py              ← load_model() + predict()
├── api/
│   └── main.py                   ← FastAPI app with Prometheus metrics
├── tests/
│   ├── conftest.py, test_data_processing.py
│   ├── test_inference.py, test_api.py
├── notebooks/
│   ├── 01_eda.ipynb              ← EDA with 6 visualisation plots
│   ├── 02_model_training.ipynb   ← Training + MLflow walkthrough
│   └── 03_inference.ipynb        ← Inference + threshold analysis
├── models/
│   ├── best_model.joblib         ← Serialised best pipeline
│   └── model_metadata.json       ← Metrics + feature schema
├── screenshots/                  ← 17 screenshots across all sections
├── k8s/deployment.yaml, service.yaml
├── monitoring/prometheus.yml
├── .github/workflows/ci-cd.yml   ← lint → test → train → docker
├── Dockerfile, docker-compose.yml
├── requirements.txt              ← Pinned dependencies
├── API_ACCESS.md                 ← Local testing instructions
└── MLOps_Assignment_Report.pdf   ← This report
```

API Access Instructions

```
# Local Python
uvicorn api.main:app --reload --port 8000

# Docker
docker run -p 8000:8000 heart-disease-api:latest
```

```
# Full stack
docker compose up --build

# Test prediction
curl -X POST http://localhost:8000/predict \
  -H "Content-Type: application/json" \
  -d '{"age":63,"sex":1,"cp":3,"trestbps":145,"chol":233,
       "fbs":1,"restecg":0,"thalach":150,"exang":0,
       "oldpeak":2.3,"slope":0,"ca":0,"thal":1}'
```
